hold light so tightly in its grip that the light cannot escape from the hole's horizon." This is how my colleagues and I as physicists think, when struggling to understand the Universe. It is a fruitful way to think; it has helped produce deep new insights into imploding stars, black holes, gravitational waves, and other phenomena.

This viewpoint is incompatible with the common view that physicists work with *theories* which try to describe the Universe, but which are only human inventions and have no real power over the Universe. The word *theory*, in fact, is so ladened with connotations of tentative-ness and human quirkiness that I shall avoid using it wherever possible. In its place I shall use the phrase *physical law* with its firm connotation of truly ruling the Universe, that is, truly forcing the Universe to behave as it does.

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1. Readers who wish to understand what is meant by "*mathematically manipulating*" the laws of physics will find a discussion in the notes section at the end of the book

2. Chapter 10.

3. This means that it was a bit unfair of me to use a high-speed sports car, which feels the Earth's gravity, in my example above. However, it turns out that because the Earth's gravitational pull is perpendicular to the direction of the car's motion (downward versus horizontal), it has no effect on any of the issues discussed in the sports-car story.

4. But see Chapter 14 for a caveat.